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Scalable Diffusion Models with Transformers

(ICCV 2023; cited by 9483)

8th Oct

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Background

Images Generated by a Diffusion Model



What Is a Diffusion Model?

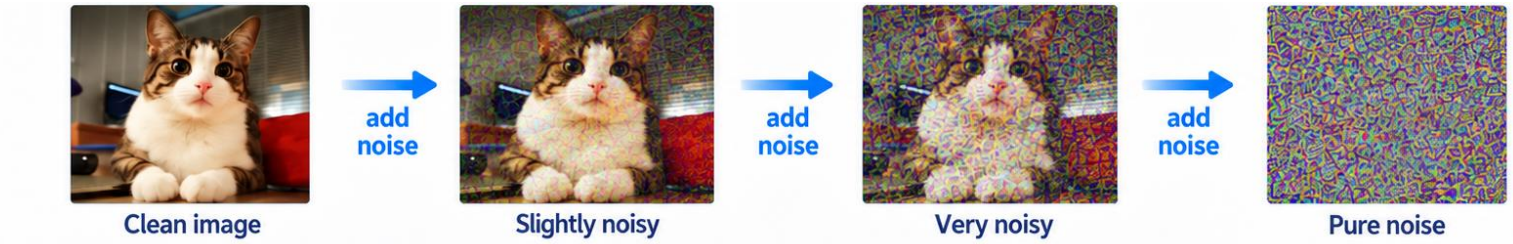
- Diffusion Model

  A model that learns to generate data by reversing a gradual noising process.

  - Gradually add noise

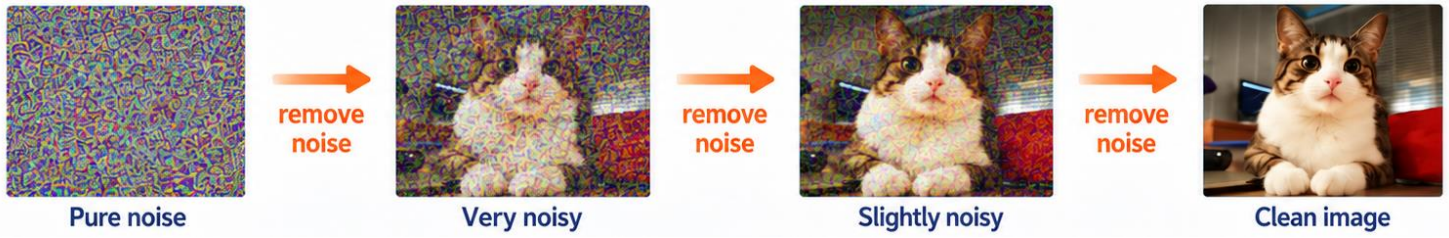
    Start from a real image;

    Add noise step by step until it becomes random noise



  - Denoising

    Train a neural network to reverse this process and generate new images



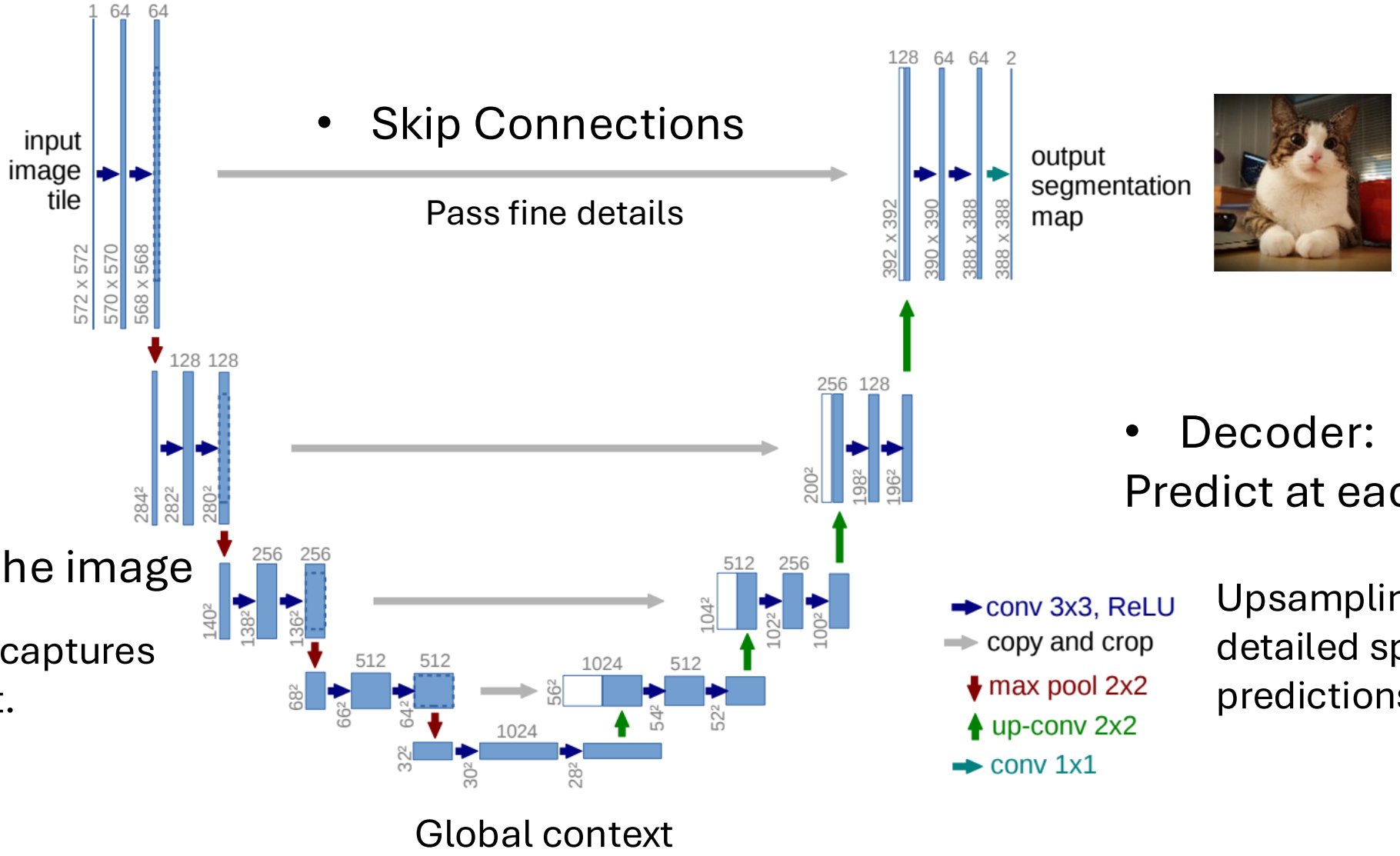
A hard problem:

"noise → image" in one step.

Many easier denoising steps.

Previous Backbone – U-Net (cited by 152k+)





- Skip Connections

Pass fine details

- Ecoder:

Understand the image

Downsampling captures broader context.

- Decoder:

Predict at each location

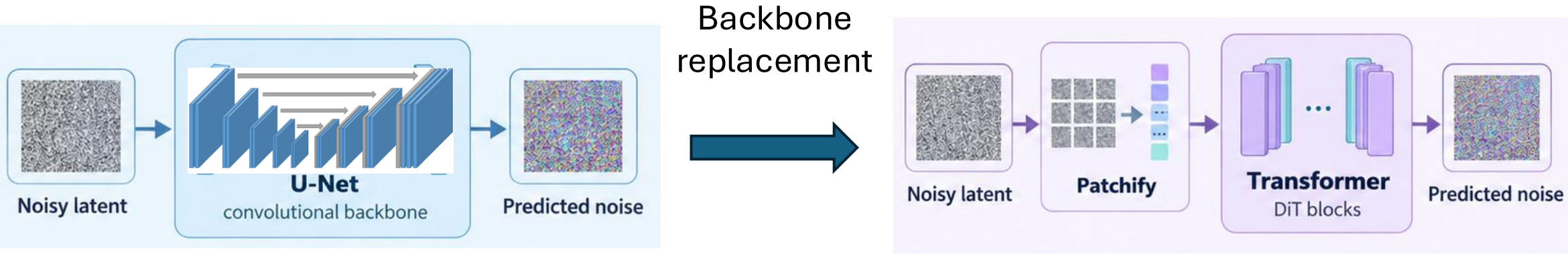
Upsampling produces detailed spatial predictions.

Global context

Methods

From U-Net to Diffusion Transformer

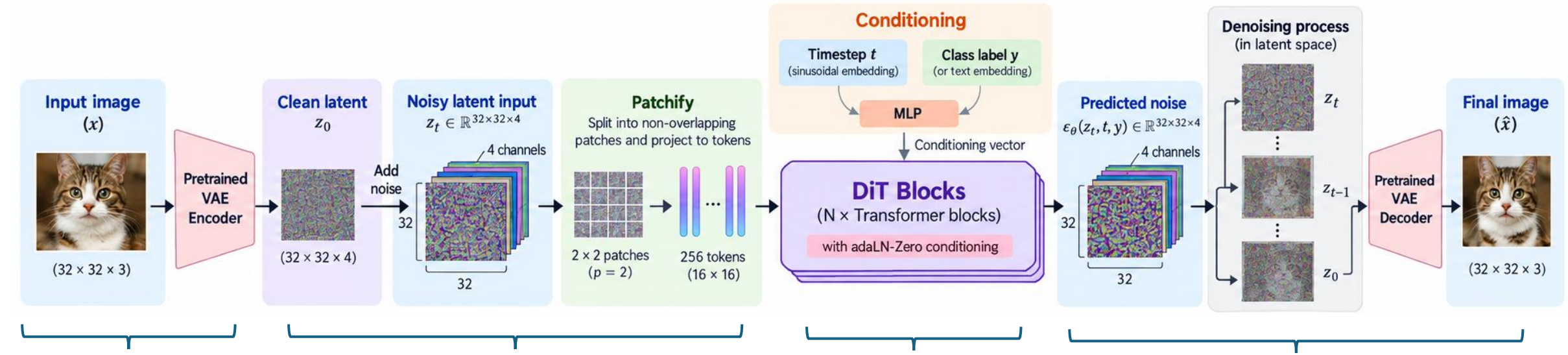
- Replace the U-Net denoiser with a scalable Transformer.



U-Net

DiT

DiT: Diffusion Transformer



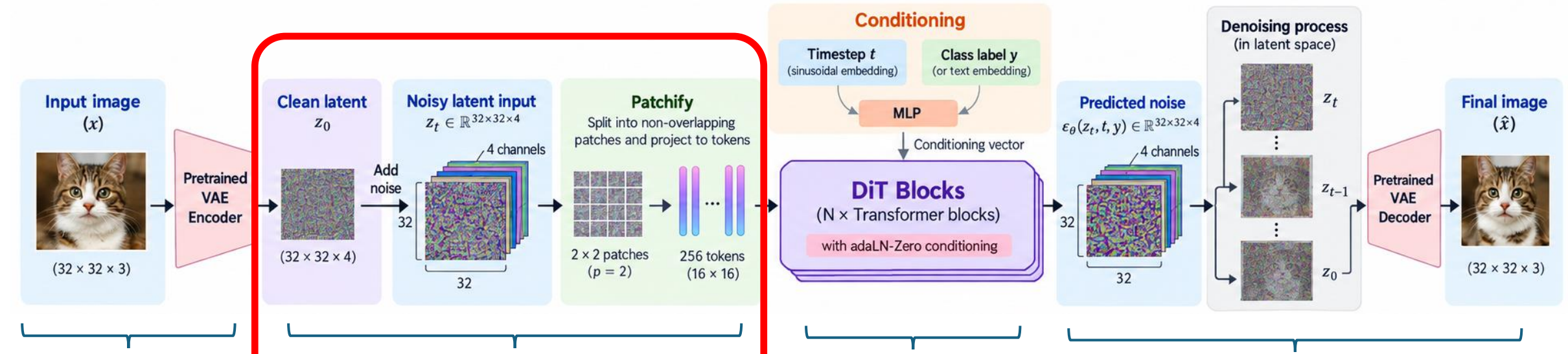
I. Input image to VAE

II. Add noise & patchify

III. DiT blocks

IV. Denoising & decode

DiT: Diffusion Transformer



I. Input image to VAE

II. Add noise & patchify

III. DiT blocks

IV. Denoising & decode

II. Add noise & Patchify

- **Add noise:** Noisy image = part of the clean image + part of random noise

controls how much original information is preserved

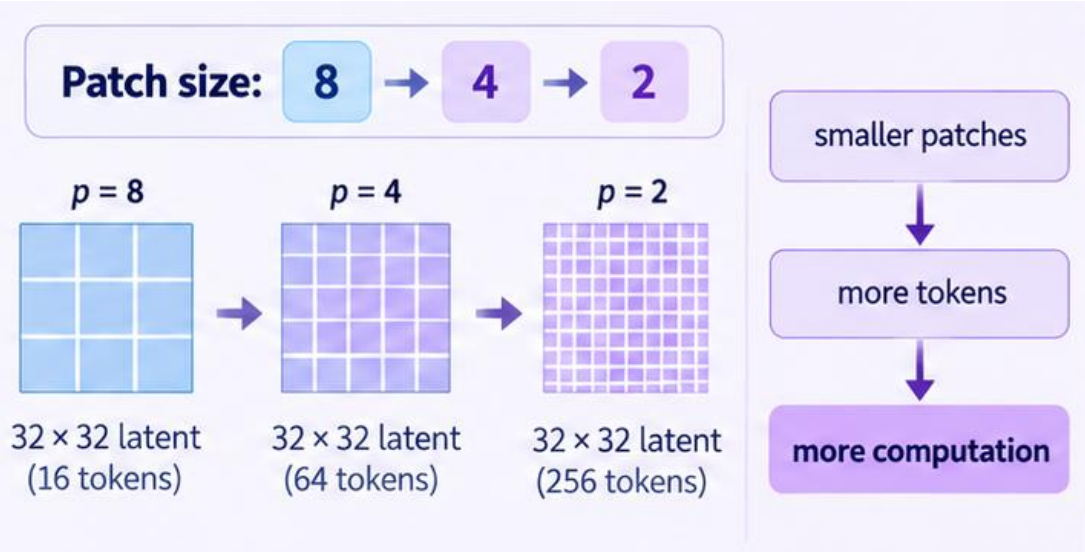
$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon_t$$

noisy image at timestep t

clean image

random Gaussian noise

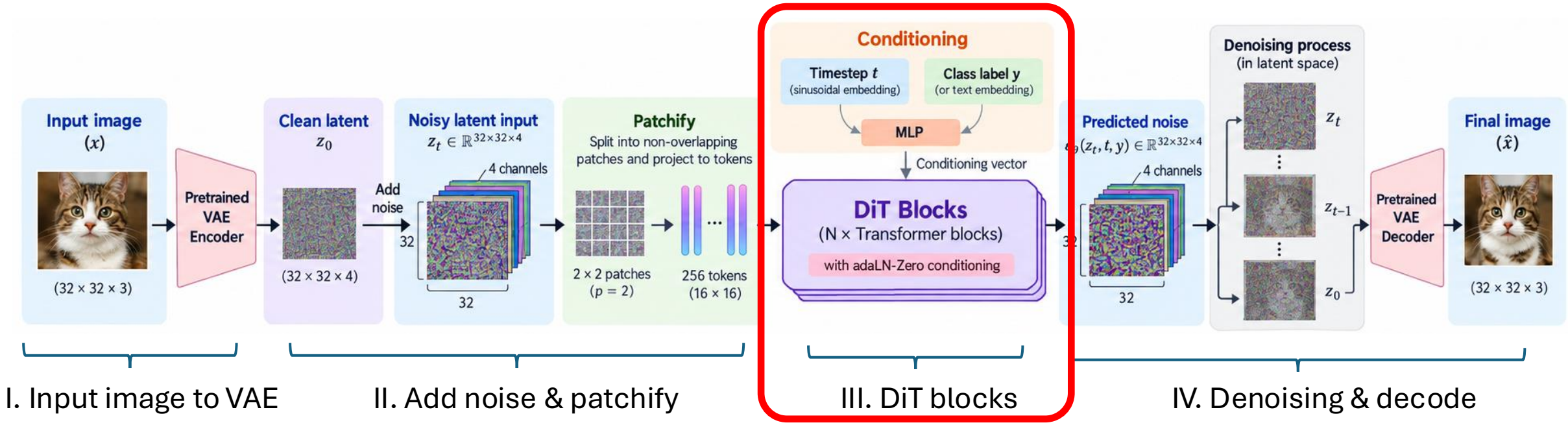
- **Patchify**: Split into non-overlapping patches and project to tokens





DiT can benefit strongly from Transformer scaling.

DiT: Diffusion Transformer



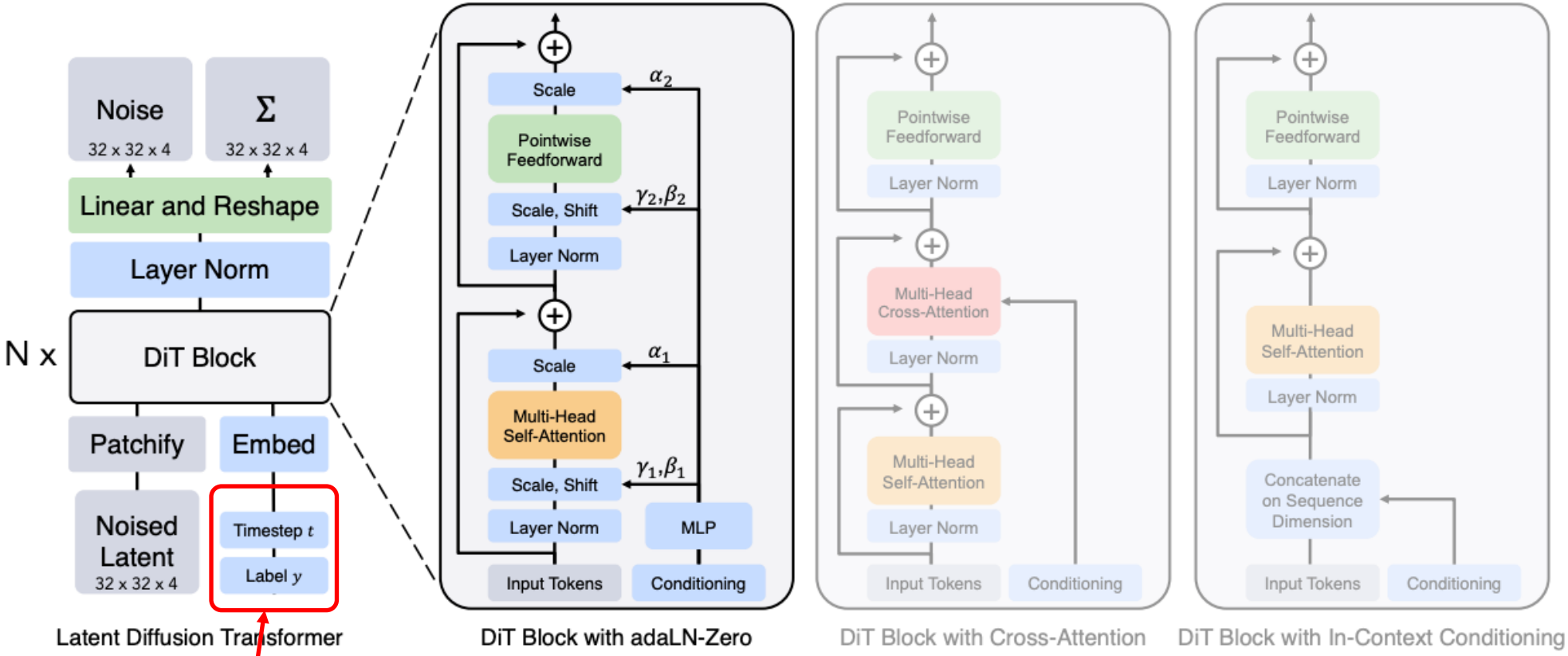
I. Input image to VAE

II. Add noise & patchify

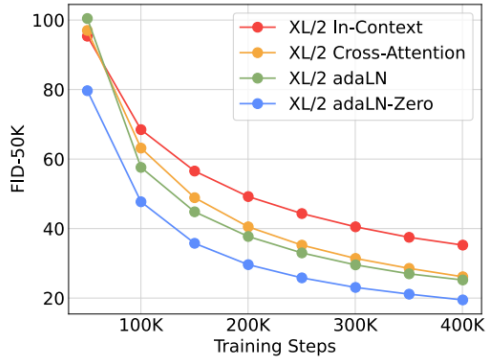
III. DiT blocks

IV. Denoising & decode

III. 3 Candidate Architectures

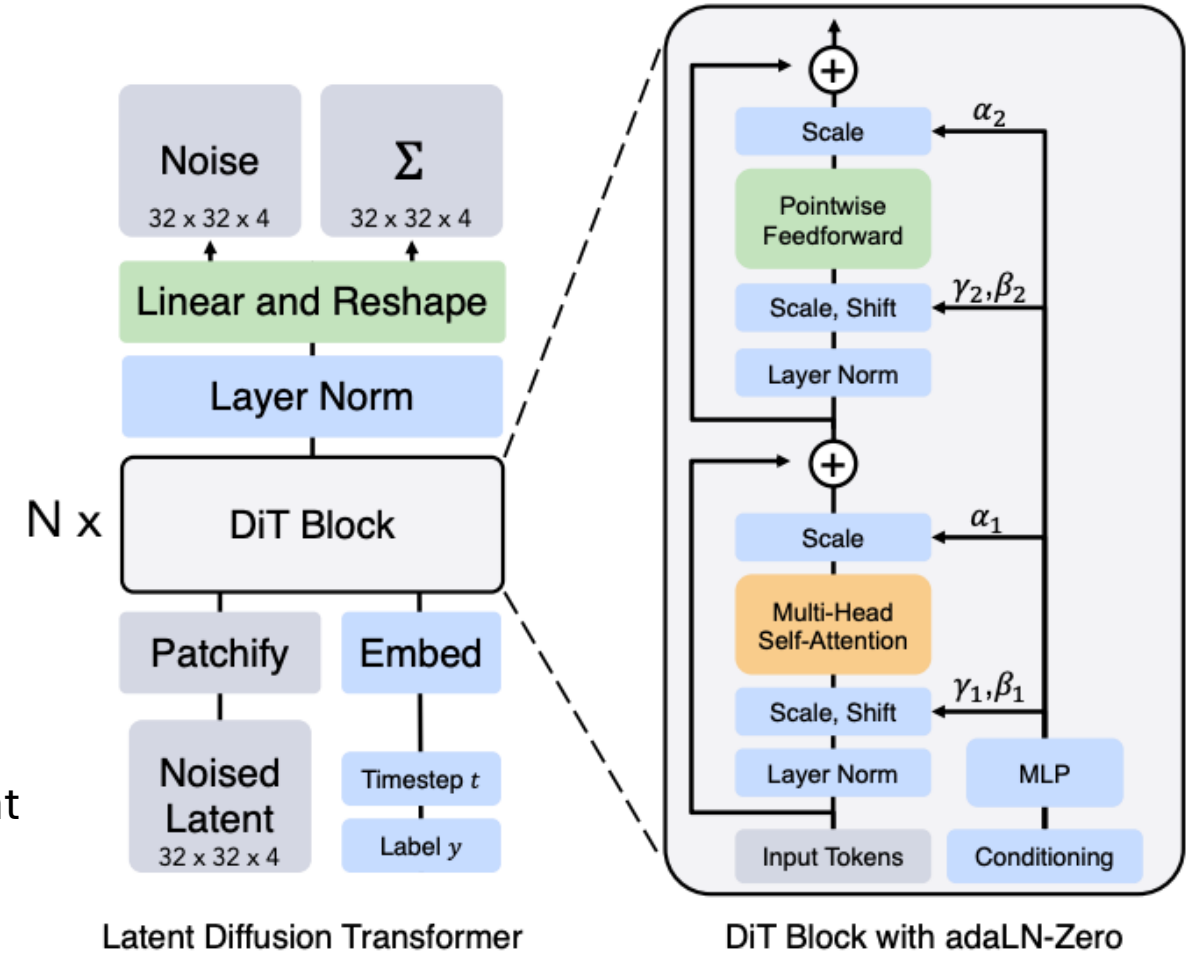


how noisy the input is,
and what it should generate



III. DiT Architecture

Output:
Predicted noise;
Variance parameters (Σ)



Scale and Shift (γ, β)

Residual Gate (α)
Control the strength of each residual branch

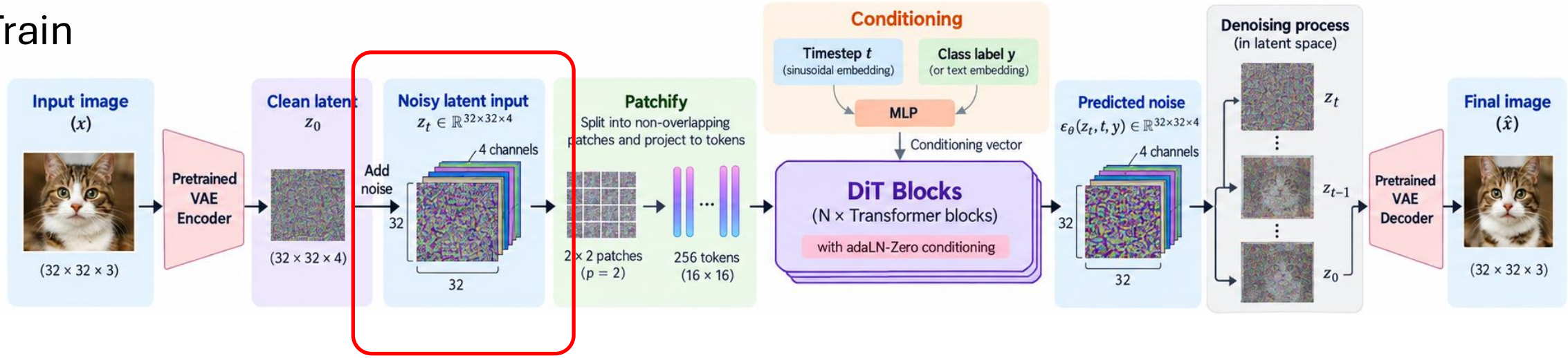
MLP produces scale, shift (γ, β), and gate parameters α.

Input

1. A noisy image representation;
2. Timestep **t**: the current noise level;
3. Class label **y**: the category to generate

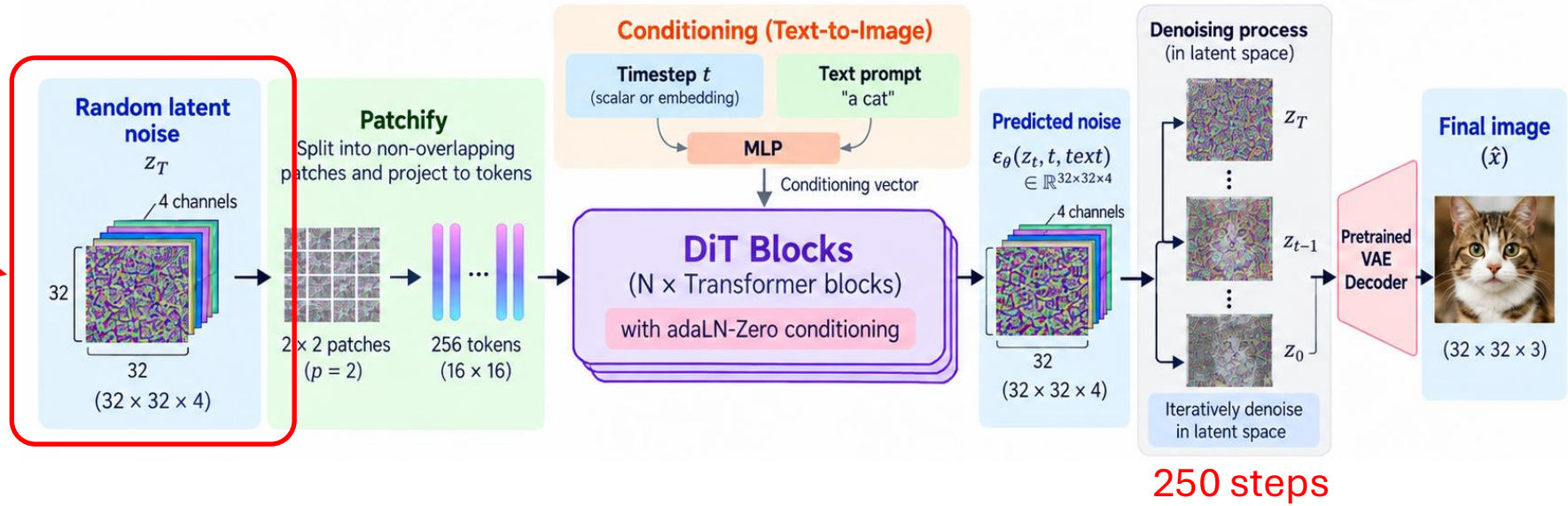
DiT: Diffusion Transformer

- Train



- Inference

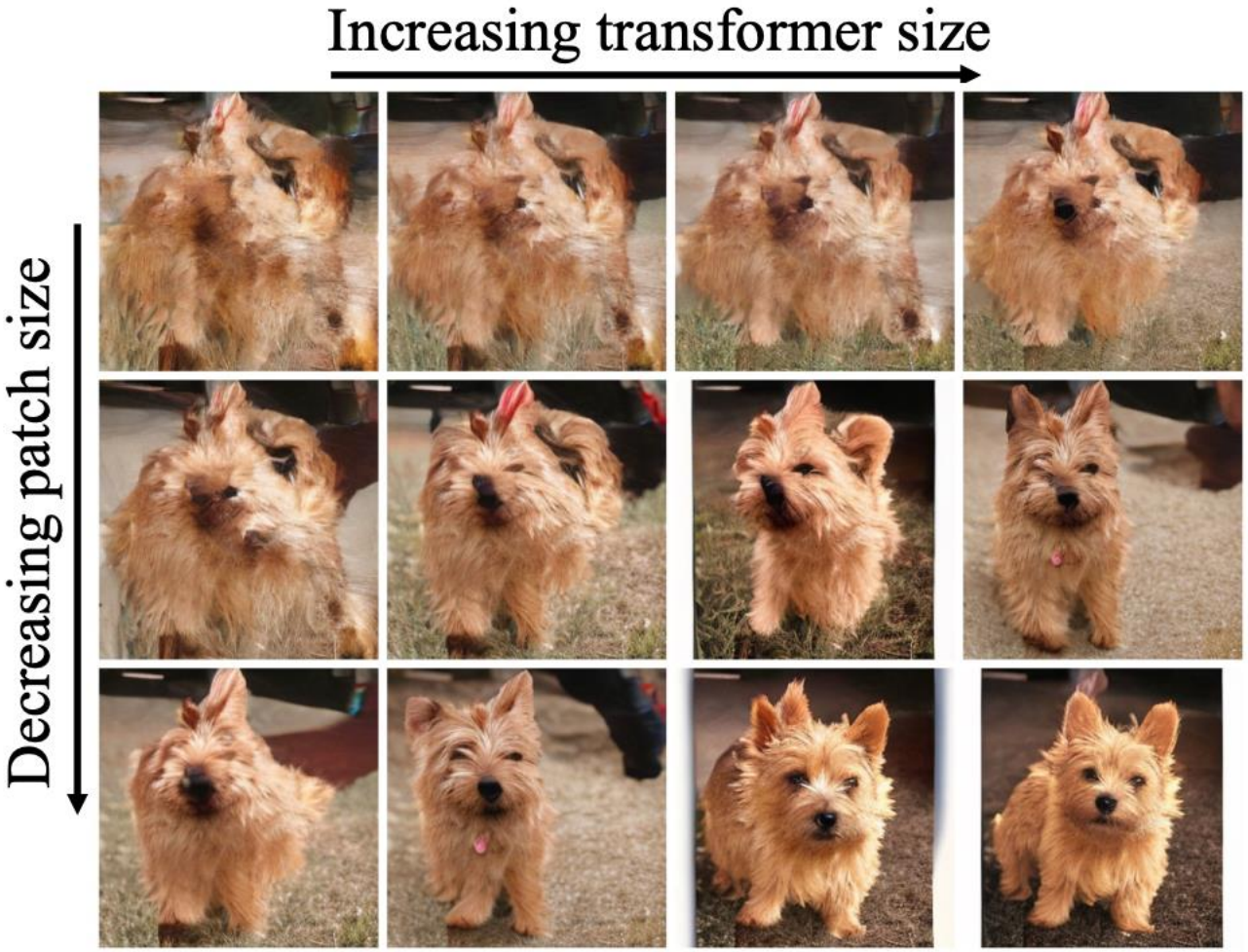
Start from random latent noise

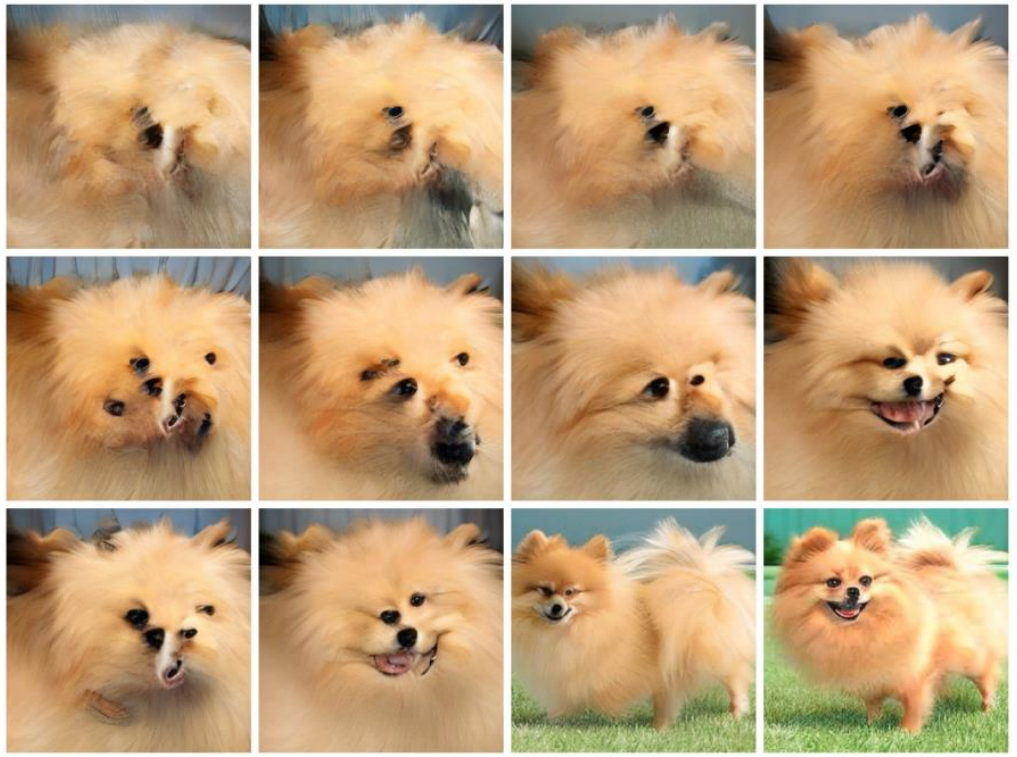


250 steps

Results

Results

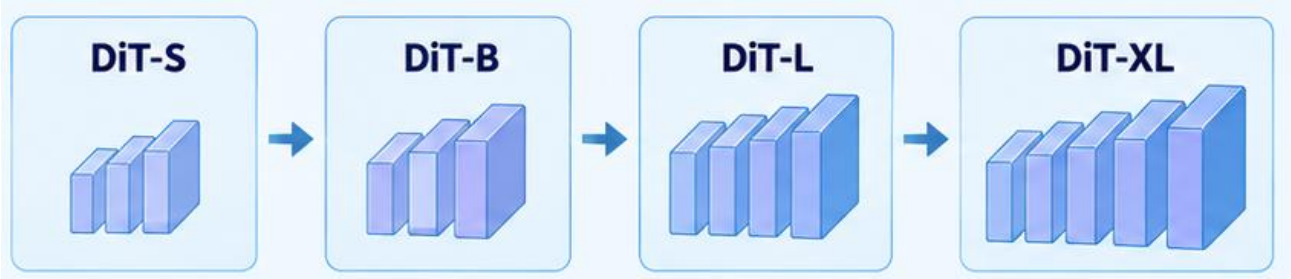


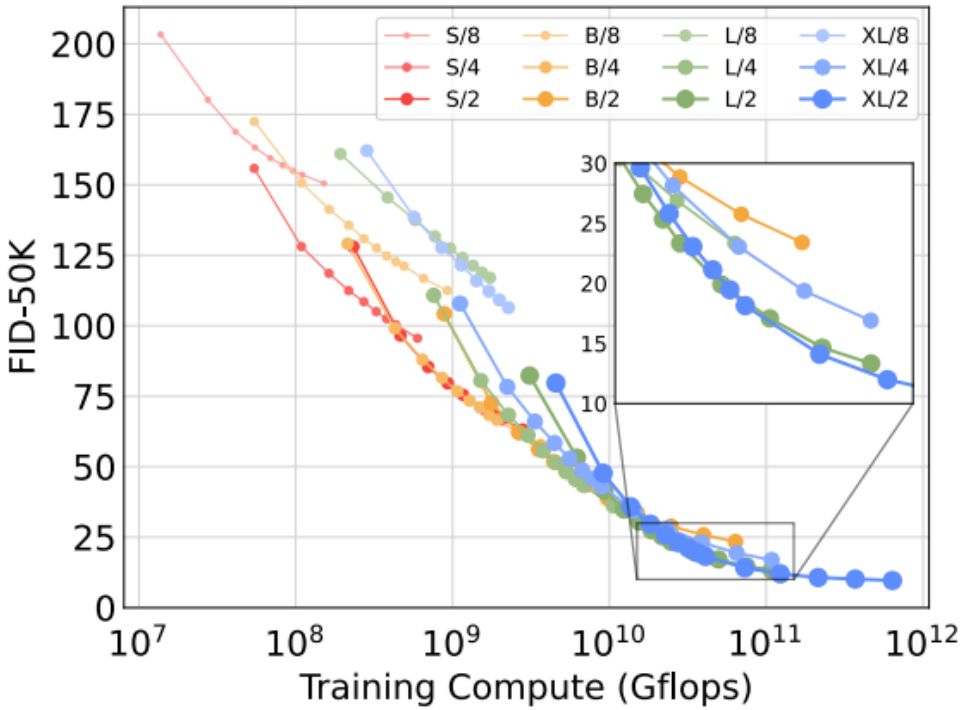


Scaling up the Transformer consistently improves diffusion model performance.

Results

- Increase model size





Conclusion

- Transformers can effectively replace U-Nets as the backbone of diffusion models.

- More compute consistently leads to better generation quality.

Thank you!

Happy to take questions.